

Simple Robot Dog - Mechanical Design Project Report

Abstract

This project presents the conceptual mechanical design of a simple quadruped robot dog. The purpose is to understand robot body structure, leg mechanisms, joints, stability, and basic engineering principles used in mobile robotics.

Introduction

Quadruped robots are widely used in research, inspection, rescue operations, agriculture, and industrial automation. Compared with wheeled robots, they can move over uneven terrain and maintain stability.

Project Objectives

Design a lightweight mechanical structure; understand robot joints and DOF; analyze stability; select suitable materials; prepare an initial engineering concept.

Mechanical Structure

The robot body consists of an aluminum chassis measuring approximately 25 cm x 15 cm x 8 cm. Four identical legs are mounted symmetrically to distribute weight evenly.

Leg Design

Each leg contains hip and knee joints driven by servo motors. The links are designed to provide sufficient movement while keeping the center of gravity close to the body.

Degrees of Freedom

The proposed educational model contains eight degrees of freedom with two active joints per leg. Increasing DOF improves flexibility but also increases cost and control complexity.

Material Selection

Aluminum is selected because it offers low weight, corrosion resistance, high strength, and easy machining. Steel fasteners improve structural reliability.

Motor Selection

MG996R servo motors are suitable for prototype development because they are inexpensive, easy to control using Arduino, and provide acceptable torque.

Stability Analysis

The battery and controller are placed near the center of the chassis to lower the center of gravity. A crawl gait improves balance because three legs remain in contact with the ground during movement.

Expected Challenges

Joint wear, motor heating, vibration, screw loosening, battery consumption, and reduced performance on slippery surfaces are expected engineering challenges.

Future Improvements

Future versions may include brushless actuators, sensors, cameras, LiDAR, force sensors, AI navigation, ROS integration, and optimized lightweight composite materials.

Conclusion

The project demonstrates the basic mechanical concepts required for designing a quadruped robot. Although simplified, it provides a strong educational foundation for future robotic development.

Technical Specifications

Specification	Value
Robot Type	Quadruped Robot
Body Length	25 cm
Body Width	15 cm
Body Height	8 cm
Legs	4
DOF	8
Material	Aluminum
Motors	8 x MG996R Servo
Estimated Weight	1.2-1.5 kg
Recommended Controller	Arduino Uno / ESP32

Prepared by: Fahad-79